

Hardware Diagnostic Report

System Maintenance & Troubleshooting Simulations

Note: As an AI, I do not possess physical hardware to dismantle. The procedures below represent a detailed simulation and standard operating procedure based on expert IT diagnostic practices.

1. RAM Reseating Procedure & Observations

Step-by-Step Procedure

- 1. Preparation:** Shut down the system entirely, flip the PSU switch to 'Off' (O), and unplug the main power cable from the wall.
- 2. Grounding:** Pressed the case power button for 5 seconds to drain residual capacitor charge. Attached an anti-static wrist strap to the unpainted metal chassis.
- 3. Access:** Removed the side panel to expose the motherboard. Located the DIMM slots adjacent to the CPU socket.
- 4. Removal:** Pushed down on the retention clips (at the top and bottom of the slot, depending on motherboard design). The RAM module popped up slightly. Gently pulled the module straight out by the edges.
- 5. Inspection & Cleaning:** Inspected the gold contacts for oxidation or debris. Used compressed air to blow out the empty DIMM slot.
- 6. Re-insertion:** Aligned the notch on the RAM stick with the key in the DIMM slot. Applied even, firm pressure on both ends of the module until the retention clips snapped back into place with an audible "click."
- 7. Verification:** Reconnected power and booted into the BIOS to ensure the memory was recognized at the correct capacity and speed.

Simulated Issues Observed

- **Dust Buildup:** Minimal dust observed in slot B2, which could have eventually bridged contacts and caused intermittent boot failures.
- **Incomplete Seating:** During initial re-insertion, one side did not fully click in. System failed to POST, emitting three long beeps (Memory Error). Pressing firmly on the unseated corner resolved the issue.

2. GPU Troubleshooting Checklist

A comprehensive 5-step checklist for diagnosing a faulty graphics card in a gaming PC:

1. **Symptom Identification:** Verify the exact nature of the failure. Check for visual artifacts (checkered patterns, green lines), black screens during heavy loads, driver timeouts, or the system failing to POST with a VGA debug light on the motherboard.
2. **Physical Connections & Reseating:** Turn off the PC. Reseat the GPU in the primary PCIe slot to ensure proper contact. Unplug and securely reconnect the PCIe power cables (6+2 pin) directly from the PSU, ensuring no splitters are overburdened. Check the display cable (HDMI/DisplayPort) on both ends.
3. **Clean Driver Installation:** Boot Windows in Safe Mode. Run Display Driver Uninstaller (DDU) to completely wipe existing GPU drivers and registry keys. Reboot and install the latest stable drivers directly from the manufacturer (NVIDIA/AMD) website.
4. **Thermals & Stress Testing:** Use software like HWiNFO64 to monitor GPU Core and Hotspot temperatures. Run a synthetic benchmark (e.g., FurMark or 3DMark Time Spy). If the PC crashes when temps exceed 85-90°C, the issue is likely thermal throttling, requiring a fan curve adjustment or repasting.
5. **Isolation & Cross-Testing (Confirming the Fix):** If issues persist, move the GPU to a secondary PCIe slot to rule out motherboard slot failure. Finally, test the GPU in an entirely different, known-working PC. If the issue follows the GPU, the hardware is confirmed faulty.

3. Storage Drive Simulation (SATA/NVMe Disconnect)

Condition: Drive Unplugged

Simulation: The primary NVMe M.2 boot drive is removed from the system.

System Response: Upon powering on, the system bypasses the usual Windows loading screen. Instead, the BIOS halts the boot process, displaying the following standard error on a black screen:

```
Reboot and Select proper Boot device
or Insert Boot Media in selected Boot device and press a key_
```

Upon entering the BIOS/UEFI, navigating to the *Storage Information* or *Boot Priority* tab reveals that the "M2_1" slot reads as **Empty / Not Present**. No Windows Boot Manager is found.

Condition: Drive Replugged

Simulation: The NVMe drive is securely re-inserted and screwed down. Power is restored.

System Response: The BIOS immediately recognizes the drive controller. The boot priority automatically detects "Windows Boot Manager (Disk 0)." The system proceeds to boot. Occasionally, Windows may trigger a brief "*Checking file system on C:*" prompt (CHKDSK) due to the previous unexpected state change, before loading the desktop normally.

4. Diagnosing Random PC Restarts

Component to Check First: RAM (Memory)

Note: While a failing Power Supply (PSU) is a prime suspect for sudden power losses, among the requested components (RAM, GPU, Storage), RAM is the most logical first check.

Reasoning

Faulty or unstable memory causes data corruption at the kernel level. When Windows detects critical memory addressing errors or invalid page faults in active system processes, it triggers a sudden reboot to protect data integrity (often accompanied by a momentary Blue Screen of Death, such as MEMORY_MANAGEMENT, that restarts too fast to read). Storage issues typically cause freezes or boot loops, and GPU issues usually result in display driver crashes (black screens) rather than hard system resets.

Diagnostic Steps for RAM

- **Disable Overclocks:** Boot into BIOS and disable XMP / DOCP / EXPO profiles. Run the memory at its base JEDEC speed to rule out instability caused by factory overclocks.
- **Software Diagnostics:** Run *Windows Memory Diagnostic* or, for a more thorough test, create a bootable USB with *MemTest86*. Let it run for at least 4 passes to detect hardware-level bit flips.
- **Physical Isolation (Process of Elimination):** If errors are found, remove all RAM sticks except one. Test the system. Swap the stick for the other one and test again. This determines whether a specific RAM stick is faulty or if the motherboard channel is defective.

- **Slot Verification:** If a stick fails in slot A2 but passes in B2, the motherboard's memory slot may be the actual point of failure.